

The empirical recurrence for OEIS A299658: recovering a conjecture that is not written down

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Abstract

OEIS A299658 records “Empirical recurrence of order 89”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 113$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 89, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 89 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A299658 is “Number of $n \times 5$ 0..1 arrays with every element equal to 0, 1, 2, 3, 4, 5 or 7 king-move adjacent elements, with upper left element zero.”. It has offset 1 and begins

16, 512, 11888, 297071, 7411398, 185302633, 4634931975, 115940148451, ...

The entry states:

Empirical recurrence of order 89 (see link above)

As of the “Last modified” line on the live entry (Feb 15 2018 14:49:58, revision 6) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 1024$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. States with the same future contribute identically to $\iota^\top M^n \tau$, so they may be merged without changing any count; merging leaves $S' = 113$ of the 1024, and it is that smaller bound the computation below uses. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 113$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S' = 226$ exact terms of the model returns order 89 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{89} c_i a(n-i),$$

c_1	23	c_{24}	115103311634488	c_{47}	80711955113511482	c_{70}	−9209996662568880
c_2	78	c_{25}	428752646700292	c_{48}	185378757871908874	c_{71}	−4953504611025285
c_3	−379	c_{26}	329656933655978	c_{49}	144380133086068640	c_{72}	337122103422548
c_4	−7413	c_{27}	−1259537455685793	c_{50}	178224322236559	c_{73}	1573361626892416
c_5	−14367	c_{28}	−3665317211105977	c_{51}	−117553340676488790	c_{74}	318047545958344
c_6	105314	c_{29}	−2141506630754647	c_{52}	−172129349078083386	c_{75}	−373042646538909
c_7	578887	c_{30}	7364205921343200	c_{53}	−124430976133225264	c_{76}	−127315276315007
c_8	397170	c_{31}	16843551928954447	c_{54}	−498137743825379	c_{77}	271057705718740
c_9	−3918548	c_{32}	8115806886570974	c_{55}	44534695644531239	c_{78}	229650789123774
c_{10}	574526	c_{33}	−20079049684157582	c_{56}	−52011387373757780	c_{79}	36102754038767
c_{11}	27679417	c_{34}	−37994198243048140	c_{57}	−117335902458308658	c_{80}	−36223688859633
c_{12}	−162753357	c_{35}	−25085254371051141	c_{58}	−34085364676207606	c_{81}	−16768342044883
c_{13}	−1339023615	c_{36}	−2321891395192961	c_{59}	77637253203654841	c_{82}	1426283027472
c_{14}	−1637414211	c_{37}	20620898039942384	c_{60}	104582919521408636	c_{83}	3363114321316
c_{15}	9670535606	c_{38}	80104945009027852	c_{61}	71101749115901218	c_{84}	833587246346
c_{16}	34187150838	c_{39}	146102792782601887	c_{62}	21722718507835496	c_{85}	−70041148466
c_{17}	14326871690	c_{40}	70594455243613700	c_{63}	−24399368099738986	c_{86}	−45379115112
c_{18}	−51379980286	c_{41}	−165110804176230644	c_{64}	−36032859143820636	c_{87}	−2810619008
c_{19}	311552991339	c_{42}	−293025345645286006	c_{65}	−10795972100167547	c_{88}	282287040
c_{20}	833298842132	c_{43}	−127146470555171519	c_{66}	14702227210031712	c_{89}	12026880
c_{21}	−4612428571412	c_{44}	104056728221711802	c_{67}	20682853272018373		
c_{22}	−24833534489488	c_{45}	115436373033475358	c_{68}	9209771518726402		
c_{23}	−26209555311164	c_{46}	26757693627740158	c_{69}	−4228754019768714		

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 89, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $89 + 4$ terms removed returns order 89 as well, so $\deg q_{\min} = 89 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 15 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once

more on the sequence with its first $89 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 89 that this sequence satisfies — and that family turns out to have exactly one member.

References

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